



GUI Based Digital Filter Design Using MATLAB

Department of Electronics and Communication Engineering

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Objectives

1. To develop an interactive MATLAB graphical interface capable of designing Lowpass, Highpass, Bandpass, and Bandstop IIR filters.
2. To compare the magnitude response of Butterworth, Chebyshev-I, Chebyshev-II and Elliptical Filter.
3. To analyze the stability of the generated filters using pole zero plot.
4. To calculate number of Adders, multipliers and delay blocks required for physical realization of filters in Direct form I and II structures.
5. To test the designed filters on noisy audio signals.

Introduction

What is an IIR Filter?

- IIR stands for **Infinite Impulse Response**.
- Unlike FIR filters, IIR filters use feedback, meaning their current output depends on current inputs, past inputs, and past outputs.
- $$y(n) = \sum_{k=0}^N b_k x(n-k) - \sum_{k=0}^N a_k y(n-k)$$
- They are highly efficient and can achieve a very steep roll-off (transition band) with a much lower filter order than FIR filters.
- It is faster than FIR filter.
- Less computational complexity hence saving memory and power.

Introduction

What is a MATLAB GUI?

- A Graphical User Interface (GUI) in MATLAB allows users to interact with algorithms using visual components like buttons, dropdowns, and plots, rather than typing code
- In this project, the GUI is built programmatically using MATLAB functions like `uifigure`, `uigridlayout`, and callbacks to create an intuitive, real-time testing environment for IIR Filter Implementation.

Method Used

1. IIR Filter Design Algorithm

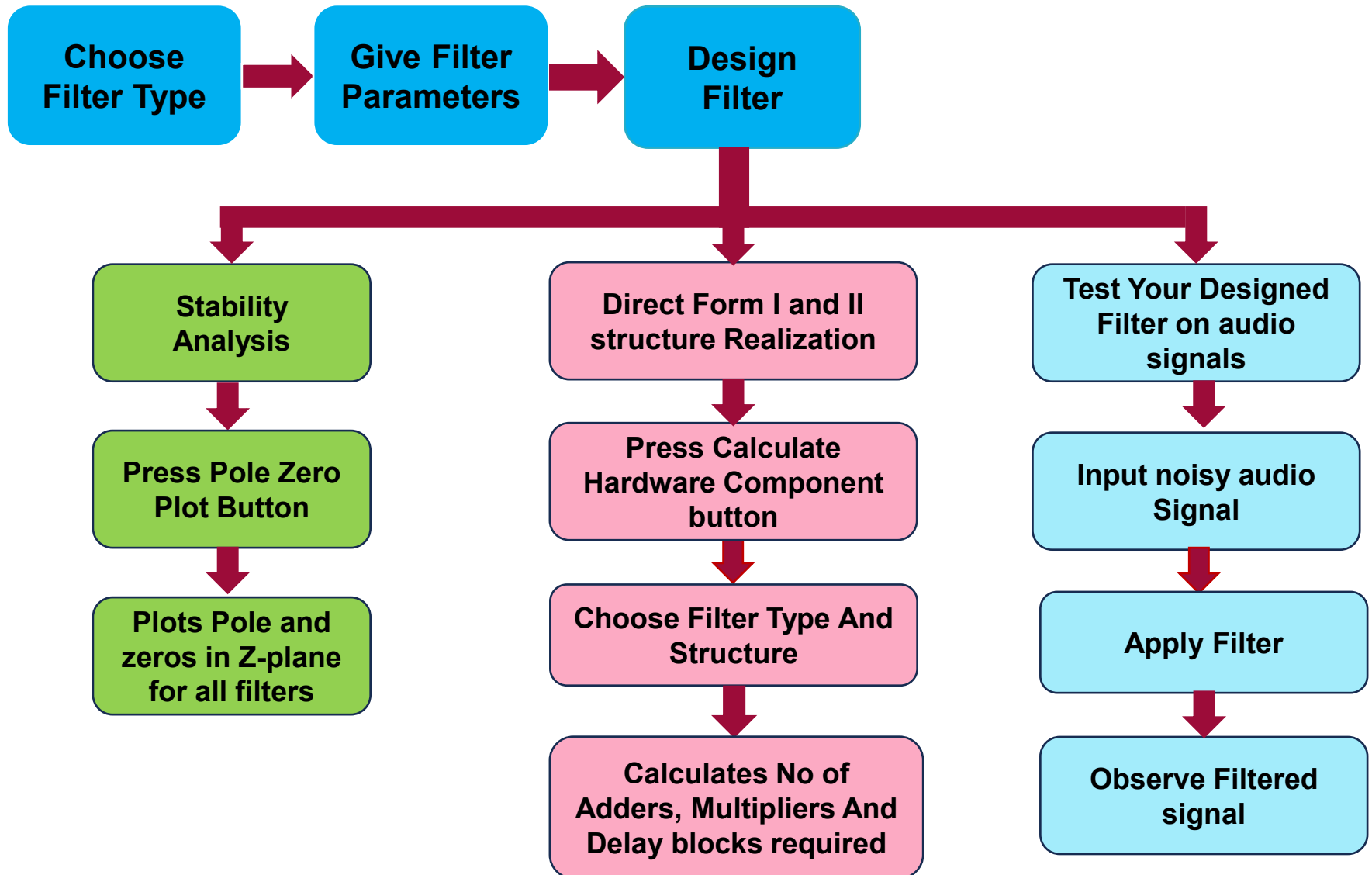
- **Parameter Extraction:** The system first captures user-defined specifications from the GUI, including Filter Order (N), Cutoff Frequency (f_c), Passband Ripple (R_p), Stopband Attenuation (A_s), and Sampling Frequency (F_s).
- **Normalized Frequency Calculation:** The absolute cutoff frequency is converted to a normalized digital frequency (ω) using the Nyquist rate.
- **Coefficient Generation:** The parameters are passed to MATLAB's core signal processing functions (`butter`, `cheby1`, `cheby2`, `ellip`) and numerator and denominator coefficients for all the filter types .

Method Used

2. GUI Implementation Flow

- **Event-Driven Architecture:** The interface is built programmatically using MATLAB's App Designer framework (uifigure, uigridlayout), ensuring a responsive, modern layout.
- **Callback Execution:** User interactions like clicking "Design Filters" or "Test Signal" triggers specific callback functions.
- **Data Flow:** Once a callback is triggered, the system extracts the current UI states, executes the underlying DSP mathematical logic, and dynamically updates the UI axes and status labels with the new results without requiring a system restart.
- **Robust Error Handling:** The application implements try-catch blocks and active input validation to prevent system crashes from invalid user entries (e.g., missing frequencies or incorrect data types), ensuring graceful failure by catching exceptions and providing immediate, clear feedback through UI alert dialogs.

Block Diagram



Results

IIR Filter design GUI Home Screen

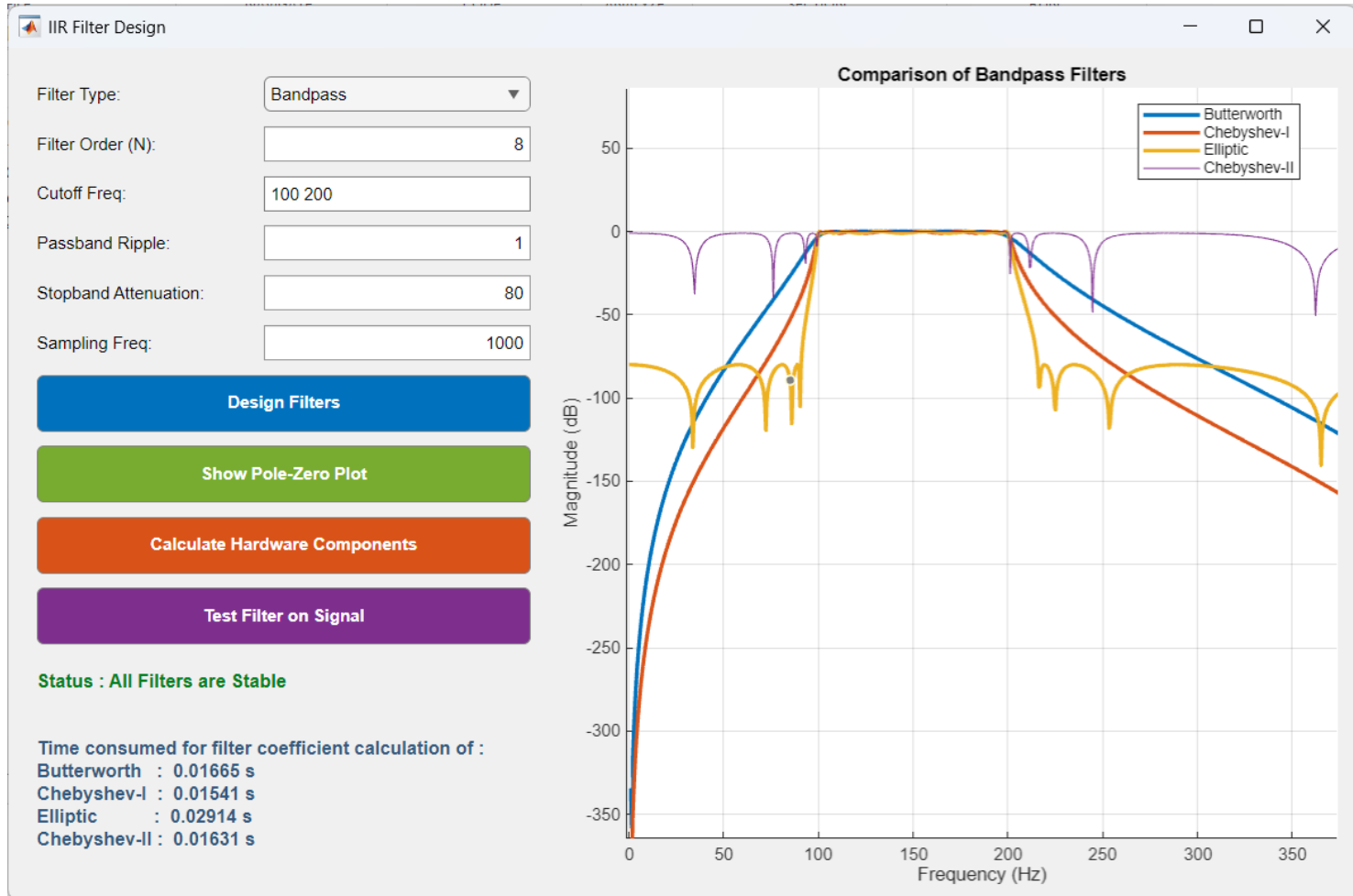
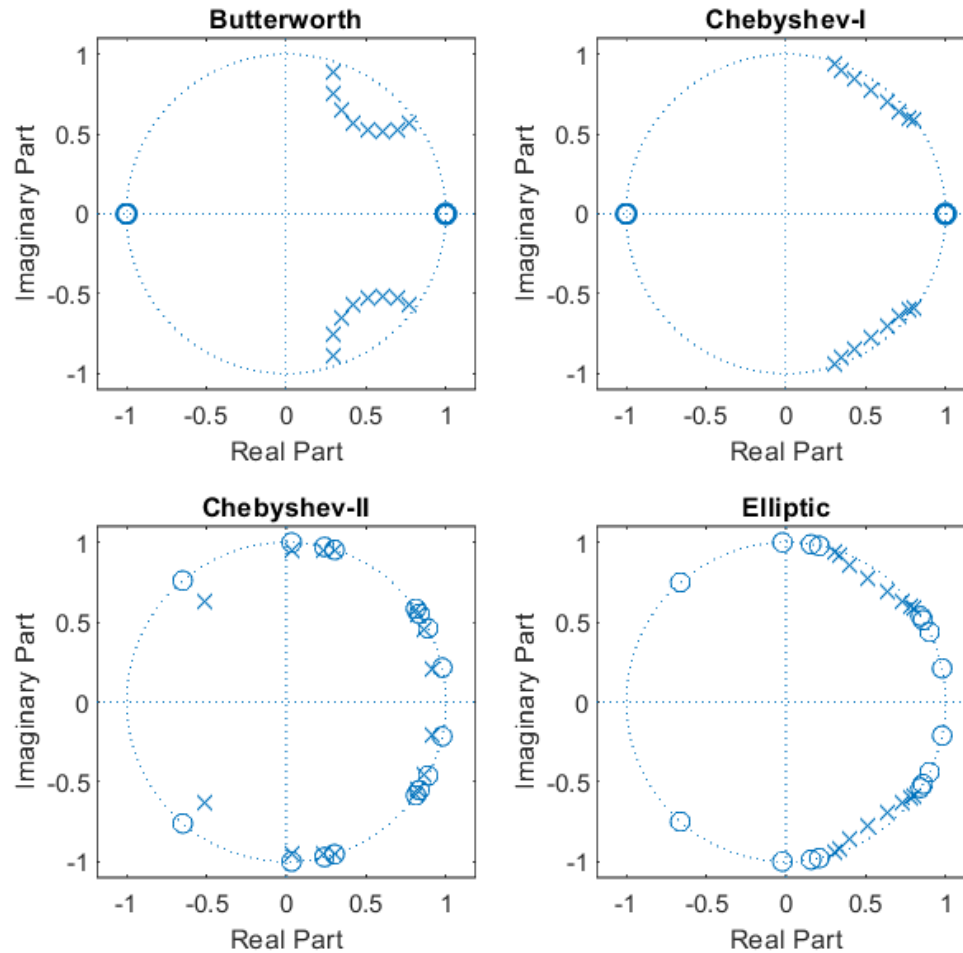


Figure shows Bandpass filter Implemented with Passband frequency 100Hz to 200Hz and filter Order 8

Results

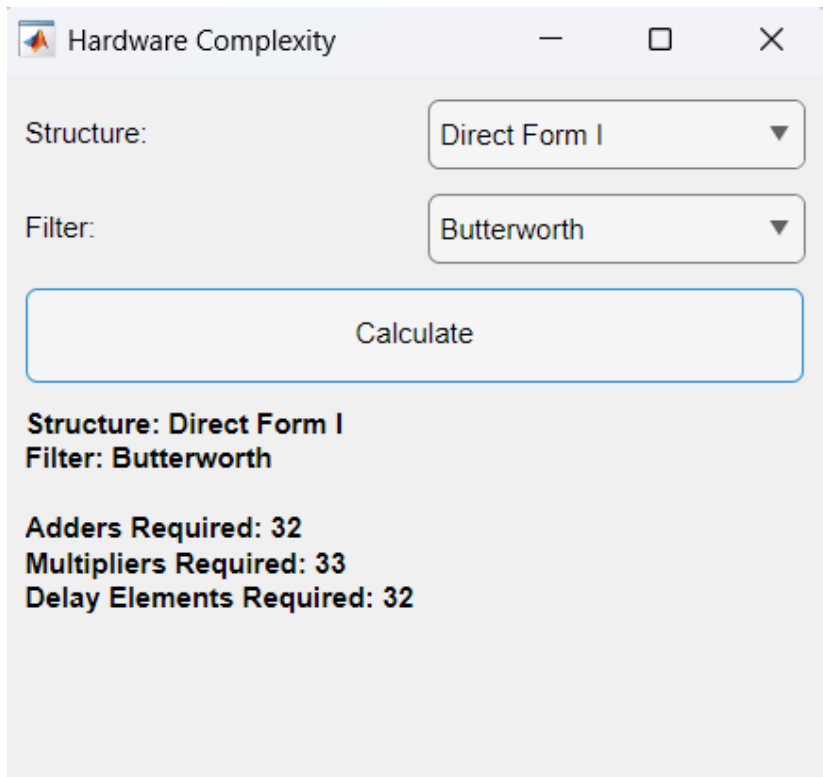
Pole Zero plot for Bandpass Filter



Pole Zero Plot for Butterworth , Chebyshev 1 and 2, Elliptic IIR Bandpass Filter

Results

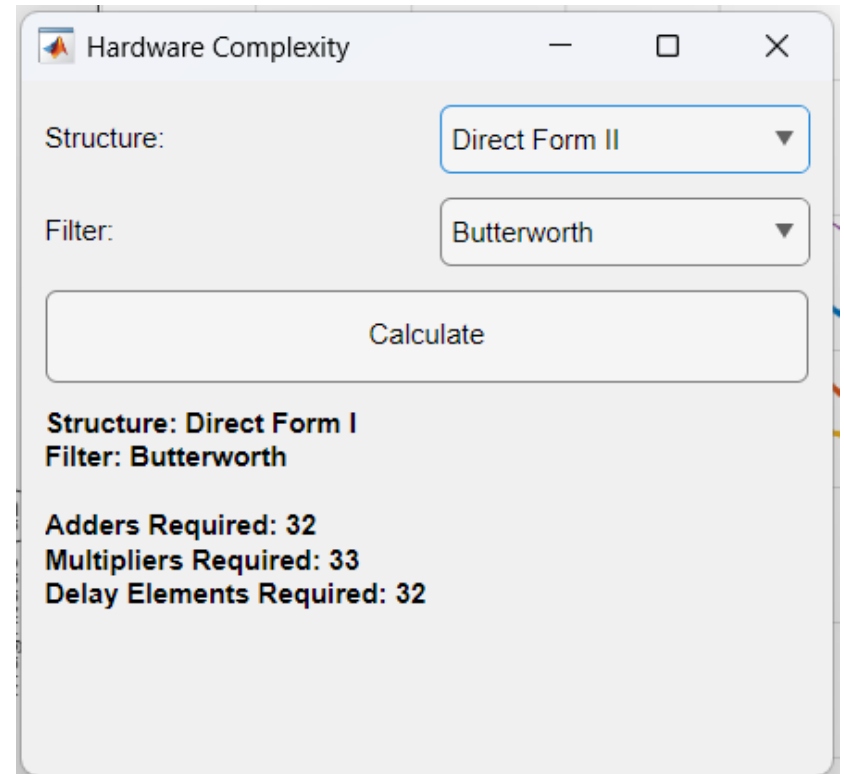
Hardware Block calculation For Direct Form I and II Realization



Hardware Complexity window showing results for Direct Form I structure. The window has a title bar with a standard icon and window controls. It contains two dropdown menus: 'Structure' set to 'Direct Form I' and 'Filter' set to 'Butterworth'. Below these is a 'Calculate' button. The results section displays the following values:

Structure: Direct Form I
Filter: Butterworth

Adders Required: 32
Multipliers Required: 33
Delay Elements Required: 32



Hardware Complexity window showing results for Direct Form II structure. The window has a title bar with a standard icon and window controls. It contains two dropdown menus: 'Structure' set to 'Direct Form II' and 'Filter' set to 'Butterworth'. Below these is a 'Calculate' button. The results section displays the following values:

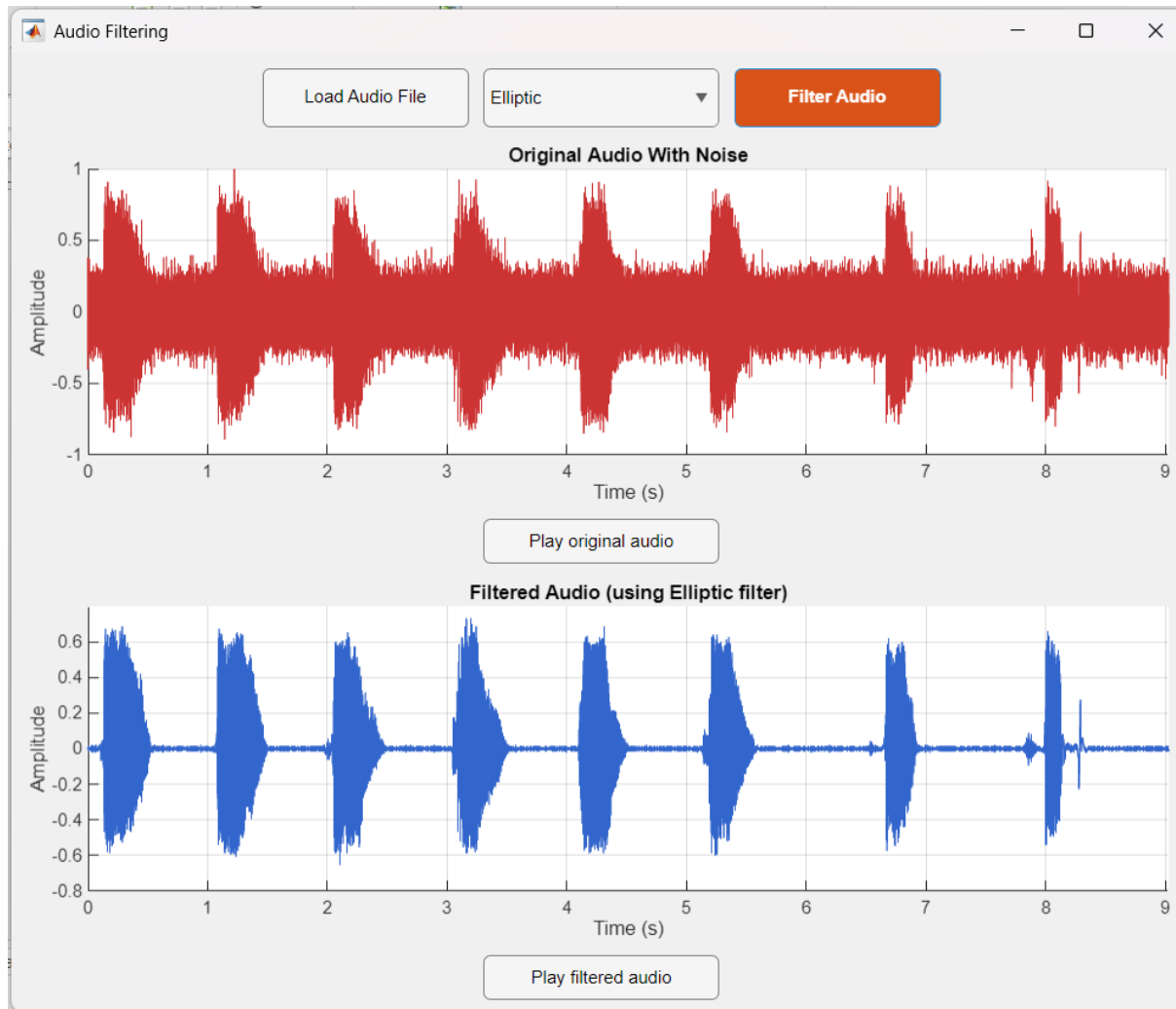
Structure: Direct Form I
Filter: Butterworth

Adders Required: 32
Multipliers Required: 33
Delay Elements Required: 32

No. of Hardware Blocks required for Direct Form I and II structure realization of Bandpass Filter with Passband 100Hz to 200Hz and Order 9

Results

Applying The designed filter on a noisy audio signal



A noisy audio signal filtered through a Lowpass Butterworth IIR Filter of Order 8 and cutoff frequency 150Hz

Results

Video

MATLAB R2024a

HOME PLOTS APPS EDITOR PUBLISH VIEW

New Open Save Compare Go To Find Refactor Analyze Run Section Break Pause Step Stop

FILE NAVIGATE CODE ANALYZE SECTION RUN

Current Folder: D:\Project\GUI Based Digital Filter design

Editor - D:\Project\GUI Based Digital Filter design\IIR_FILTER_GUI.m

```
1 function IIR_FILTER_GUI
2 % Name: Mahesh Kumar Sahoo
3 % Reg No: 2341019194
4
5 clc;clear;close all;
6
7 % Shared variables to store coefficients for the Pole-Zero plot and Filtering
8 b1 = []; a1 = [];
9 b2 = []; a2 = [];
10 b3 = []; a3 = [];
11 b4 = []; a4 = [];
12 filtersDesigned = false; % Flag to check if filters are generated
13
14 % MAIN WINDOW
15 fig = uifigure('Name', 'IIR Filter Design','Position', [100 100 950 600]);
16
17 % MAIN GRID
18 main_grid = uigridlayout(fig, [1, 2]);
19 main_grid.ColumnWidth = {'1x', '1.5x'};
20
21 % LEFT PANEL
22 left_grid = uigridlayout(main_grid, [15, 2]); % Increased rows to 13
23 left_grid.RowHeight = {25,25,25,25,25,40,40,40,40,30,30,'1x'}; % Added height for new button
24 left_grid.ColumnWidth = {150,'1x'};
25 left_grid.Padding = [10 10 10 10];
26
27 % FILTER TYPE
28 uilabel(left_grid, 'Text', 'Filter Type');
```

Workspace

Name	Value
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Command Window

Select a file to view details

Zoom: 100% UTF-8 CRLF IIR_FILTER_GUI Ln 66 Col 5

Advantage

- **Instant Comparative Analysis:**

The tool calculates and plots the magnitude responses of four different IIR filter families (Butterworth, Chebyshev I & II, and Elliptic) on the same graph simultaneously, allowing for immediate visual comparison of trade-offs like roll-off steepness and passband ripple.

- **Automated Stability Verification:**

The application automatically maps the complex Z-plane and executes mathematical checks to flag unstable filters, preventing system failures.

- **Robust and User-Friendly:**

The use of modern UI components (uigridlayout, uieditfield) paired with comprehensive try-catch error handling ensures that the application responds gracefully to incorrect inputs (like missing frequencies) without crashing.

- **All-in-One Integrated Environment:**

This GUI consolidates filter design, stability analysis, hardware estimation, and real-world signal filtering using designed filter.

Applications

1. Control Systems

- **PID Controller Tuning**
- **Vibration Damping**

2. Embedded Systems

- **Power Efficiency:** Because IIR filters require 1/10th the mathematical operations of an FIR filter, they consume significantly less CPU time, dramatically extending battery life in IoT devices.
- **Smart Sensors:** Running continuous real-time low-pass filtering on tiny IMU chips

3. Real-Time Systems

- **Bio-Signal Processing:** Smoothing noisy ECG/EEG data for immediate diagnostic feedback
- **Anti-Aliasing:** Acting as a pre-filter to block high-frequency noise right before Analog-to-Digital Conversion (ADC).
- **Active Noise Cancellation (ANC)**
- **Echo Cancellation:** Deploying adaptive IIR filters to predict and subtract delayed voice reflections during live video or audio calls.

Applications

Filter Selection by Application

- **Audio Telephony:** Uses **Elliptic** filters because they provide the sharpest possible cut-off to tightly band-limit voice signals and save bandwidth.
- **Bio-Medical (ECG):** Uses **Butterworth** filters because their perfectly smooth passband preserves the true, physical shape of the heart's electrical waveform.
- **Real-Time Control:** Uses **Chebyshev Type II** because the flat passband prevents any distortion of the primary control signal inside the loop.

Conclusion

- This project successfully bridges the gap between theoretical Digital Signal Processing mathematics and practical application.
- The interactive GUI simplifies complex filter design, making it immediately visible how altering parameters like "Ripple" or "Order" impacts the final system.
- By integrating Pole-Zero plots, hardware calculators, and active filter testing into a single application, it provides a comprehensive tool for evaluating the trade-offs between different IIR filter families prior to physical implementation.

References

- [1] *MatLab Based Design and Implementation of Digital Filter* by Zena Ez Dallalbashi (IJCSNS, Vol. 20 No. 1, January 2020).
- [2] The MathWorks Inc., "Create Apps with App Designer," MathWorks Documentation, 2023.
- [3] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th ed. Upper Saddle River, NJ, USA: Pearson Prentice Hall, 2006.

THANK YOU